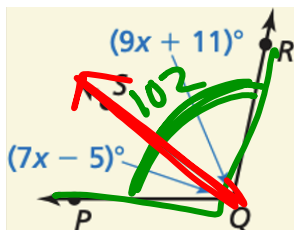


Students will describe, identify, and use the properties of angle pairs to solve problems.

Do Now

- 1) $m\angle PQR = 102^\circ$. Find the measure of the unknown angles.



$$9x + 11 + 7x - 5 = 102$$

$$16x + 6 = 102$$

$$\frac{16x}{16} = \frac{96}{16}$$

$$x = 6$$

$$9(6) + 11 = 65^\circ$$

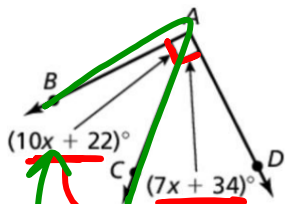
$$7(6) - 5 = 37^\circ$$

- 2) Is $\overrightarrow{QS}$ an angle bisector? Explain how you know.

No - because $m\angle PQS$ and $m\angle SQR$ are not equal.

Do Next

- 3) Find $m\angle BAC$.

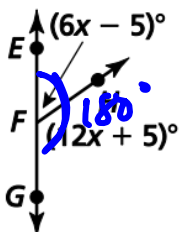


$$90 = 10x + 22 + 7x + 34$$

$$90 = 17x + 56$$

$$x = 2$$

- 4) $\angle EFG$ is a straight angle. Find $m\angle HFG$.



$$6x - 5 + 12x + 5 = 180$$

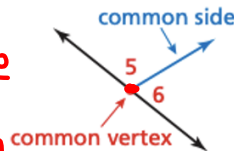
Questions on Homework?

Describing Pairs of Angles

It can often be very helpful to describe relationships between two angles. The following vocabulary will be helpful in those descriptions:

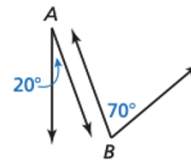
Adjacent Angles:

2 $\angle$ s that are next to each other; they must share a common side and common vertex.



Complementary Angles:

2 $\angle$ s that add to 90° . Ex: " $\angle A$ and $\angle B$ are complementary."
 $20 + 70 = 90$



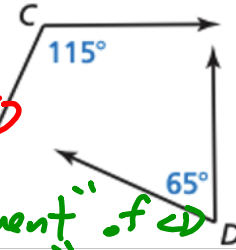
- $\angle A$ is "the complement" of $\angle B$
- $\angle B$ is "the complement" of $\angle A$
- 40 is the complement of 50.

Supplementary Angles:

2 $\angle$ s that add to 180° .

Ex: $\angle C$ and $\angle D$ are supplementary
because $65 + 115 = 180^\circ$

- $\angle C$ is "the supplement" of $\angle D$
- $\angle D$ is "the supplement" of $\angle C$
- 70 is the supplement of 110 .



Ex 1: For each image below, name a pair of adjacent angles, a pair of complementary angles, and a pair of supplementary angles.

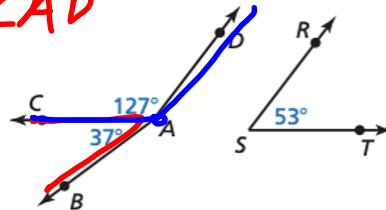
Adjacent angles: $\angle CAB$ and $\angle CAD$

Complementary angles:

$\angle CAB$ and $\angle RST$

Supplementary angles:

$\angle CAD$ and $\angle RST$



Practice

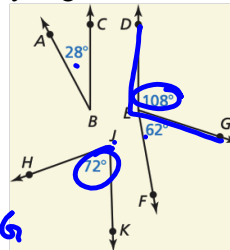
1) For each image below, name a pair of adjacent angles, a pair of complementary angles, and a pair of supplementary angles.

Adjacent angles: $\angle DEG$ and $\angle FEG$

Complementary angles:

$\angle ABX$ and $\angle GEF$

Supplementary angles: $\angle KJH$ and $\angle DEG$



2) If $\angle 1$ and $\angle 2$ are complementary angles and $m\angle 1 = 62^\circ$, find $m\angle 2$.

$$m\angle 1 + m\angle 2 = 90$$

$$62 + m\angle 2 = 90$$

$$-62$$

$$-62$$

$$m\angle 2 = 28^\circ$$

3) $\angle LMN$ and $\angle PQR$ are complementary angles. $m\angle LMN = (4x - 2)^\circ$ and $m\angle PQR = (9x + 1)^\circ$.

a. Set up an equation that represents the relationship in the situation.

$$4x - 2 + 9x + 1 = 90$$

b. Solve for x .

$$13x - 1 = 90$$

c. Find $m\angle LMN$.

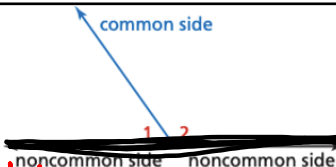
$$\frac{13x}{13} = \frac{91}{13}$$

d. $m\angle PQR$.

$$x = 7$$

Linear Pairs:

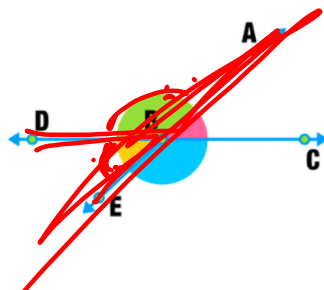
2 adjacent $\angle$ s
that have non-common
sides that form a straight $\angle$.



Important fact:

Ex: " $\angle 1$ and $\angle 2$ are a linear pair"
"Linear pairs add to 180° "
"Linear pairs are supplementary"

4) List all the linear pairs shown in the diagram below.



$\angle ABC$ & $\angle DBA$

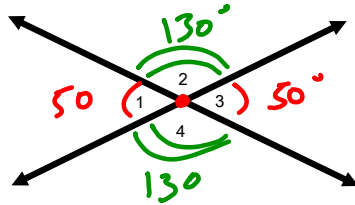
$\angle ABC$ & $\angle EBC$

$\angle DBE$ & $\angle EBC$

, $\angle DBA$ & $\angle DBE$

Vertical Angles

Vertical angles are the angles that are opposite each other when two lines intersect.



$\angle 1$ & $\angle 3$ are vertical angles.

$\angle 2$ & $\angle 4$ are vertical angles.

Vertical angles are always congruent!

$$\angle 1 \cong \angle 3$$

$$m\angle 1 = m\angle 3$$

and

$$\angle 2 \cong \angle 4$$

$$m\angle 2 = m\angle 4$$

If $m\angle 2 = 130^\circ$, can you find the measures of all the other angles?

see diagram

Practice

5) Lines l and m intersect as shown below. Determine whether the angle pairs shown below are linear pairs, vertical angles, are neither.

a. $\angle 1$ and $\angle 4$

Linear Pair

b. $\angle 2$ and $\angle 5$

Neither

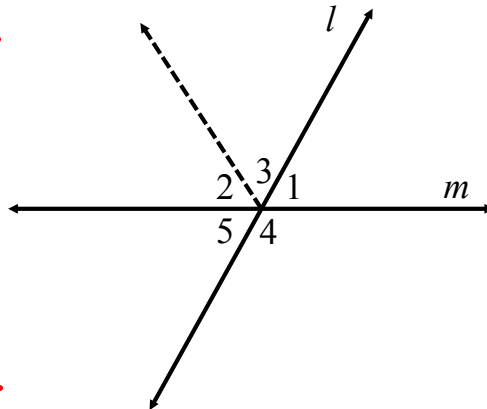
c. $\angle 1$ and $\angle 5$

Vertical $\angle$ s

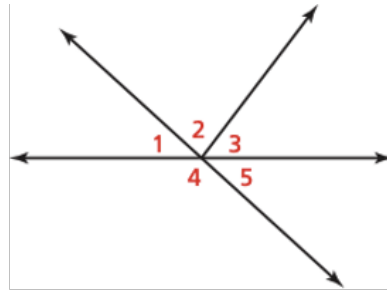
d. $\angle 4$ and $\angle 5$

Linear Pair

e. If $\angle 2 \cong \angle 3$ and $m\angle 1 = 50^\circ$, find $m\angle 3$. = 65°



6) Identify all the linear pairs and all the vertical angles in the diagram:



7) Two angles are complementary. The measure of the smaller angle is 36° less than the measure of the larger angle. Find the measure of each angle.

8) Two angles form a linear pair. The measure of the larger angle is 8 times the measure of the smaller angle. Find the measure of each angle.

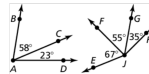
Geometry

Name: _____ Hr: _____

Assignment 1.9: Describing Pairs of Angles

In Exercises 1–3, use the figures.

1. Name a pair of adjacent complementary angles.
2. Name a pair of nonadjacent complementary angles.
3. Name a pair of nonadjacent supplementary angles.



In Exercises 4 and 5, find the angle measure.

4. $\angle 1$ is a complement of $\angle 2$, and $m\angle 2 = 36^\circ$. Find $m\angle 1$.
5. $\angle 3$ is a supplement of $\angle 4$, and $m\angle 4 = 75^\circ$. Find $m\angle 3$.

In Exercises 6 and 7, find the measure of each angle.

6. $\angle WXY$ and $\angle YXZ$ are supplementary angles, $m\angle WXY = (6x + 59)^\circ$, and $m\angle YXZ = (3x - 14)^\circ$.

a) Draw a picture matching the description.

b) Set up a relationship representing the description and picture you created.

c) Solve for x .

d) Find $m\angle WXY$ and $m\angle YXZ$.

7. $\angle ABC$ and $\angle CBD$ are complementary angles, $m\angle ABC = (3x + 6)^\circ$, and $m\angle CBD = (4x - 14)^\circ$.

a) Draw a picture matching the description.

b) Set up a relationship representing the description and picture you created.

c) Solve for x .

d) Find $m\angle WXY$ and $m\angle YXZ$.

In Exercises 8–10, use the figure.

8. Identify the linear pairs that include $\angle 5$.
9. Are $\angle 3$ and $\angle 5$ vertical angles? Explain your reasoning.
10. Are $\angle 2$ and $\angle 4$ vertical angles? Explain your reasoning.



In Exercises 11–13, write and solve an algebraic equation to find the measure of each angle based on the given description.

11. Two angles form a linear pair. The larger angle's measure is 24° more than the measure of the smaller angle.
12. Two angles are complementary. The larger angle's measure is three times the measurement of the smaller angle.
13. Two angles are supplementary. The smaller angle's measure is 15 less than half the measurement of the larger angle.

14. The figure shows the design on an outdoor fence.

- a. Name a pair of adjacent supplementary angles.
- b. Name a pair of nonadjacent supplementary angles.
- c. Identify the linear pairs that include $\angle 5$.
- d. Find $m\angle 3$. Explain your reasoning.

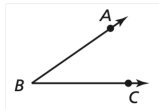


Geometry

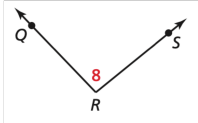
Name: _____ Hr: _____

Assignment 1.8: Angle Terminology

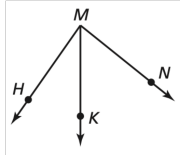
- 1) Write three names for the angle shown below:



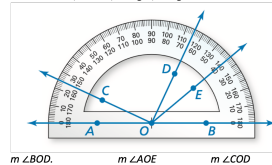
- 2) Write three names for the angle shown below:



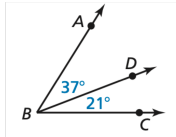
- 3) Name three different angles in the diagram:



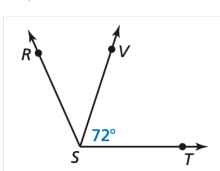
- 4) Find the measure of each angle, then classify it as acute, obtuse, straight, or right.



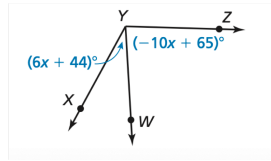
- 5) Find $m\angle ABC$



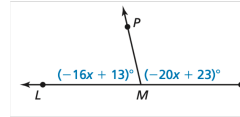
- 6) $m\angle RST = 114^\circ$. Find $m\angle RSV$.



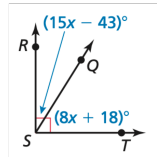
7) $m\angle XYZ = 117^\circ$. Find $m\angle XYW$ and $m\angle WYZ$.



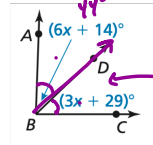
8) $\angle LMN$ is a straight angle. Find $m\angle LMP$ and $m\angle NMP$.



9) Find $m\angle RSQ$ and $m\angle TSQ$.



10) Ray BD bisects $\angle ABC$. Find $m\angle ABD$, $m\angle CBD$, and $m\angle ABC$.



$$\begin{aligned}
 6x + 14 &= 3x + 29 \\
 -3x & \quad -3x \\
 3x + 14 &= 29 \\
 -14 & \quad -14 \\
 3x &= 15 \\
 x &= 5
 \end{aligned}$$